Cloud Computing & DevOps – AWS Handbook

SESSION 8

Docker Storage, Networking & Multi-Container Applications

Building Real-World Containerized Applications

Duration	2 Hours
Prerequisites	Docker Fundamentals, Linux Basics
Learning Outcome	Understand Docker volumes, networking, data persistence and communication between multiple containers.

Why Are We Learning This?

Containers are designed to be temporary. If a container is deleted, its internal data is lost. Real-world applications require persistent storage and communication between services such as web servers, databases, and APIs. Docker volumes and Docker networks solve these challenges.

Architecture Diagram

```
Web Container
↔
Docker Network
↔
Database Container

Both Containers
↓
Docker Volume
↓
Persistent Storage
```

Key Concepts

Docker Volume	Persistent storage for containers.
Docker Network	Allows communication between containers.
Data Persistence	Retaining data beyond container lifecycle.
Bridge Network	Default Docker networking mode.
Container Communication	Interaction between multiple containers.
Multi-Container App	Application built using multiple services.

Practical Activities

• Create Docker volumes

• Attach volumes to containers

• Verify data persistence after container restart

• Create Docker networks

• Connect multiple containers to the same network

• Verify container-to-container communication

• Deploy a simple multi-container application

Expected Outputs

Docker volumes created successfully, persistent data retained after container recreation, containers communicating over Docker networks, and multi-container application functioning correctly.

What You Should See

Paste Screenshot Here Docker Volume Creation
Paste Screenshot Here Volume Verification
Paste Screenshot Here Docker Network Creation
Paste Screenshot Here Connected Containers
Paste Screenshot Here Multi-Container Application Output

Journal Writing Template

Aim

To understand Docker storage and networking concepts using volumes and multi-container applications.

Theory

Docker volumes provide persistent storage while Docker networks enable communication between containers.

Procedure

Write 5–7 concise steps performed during the practical session.

Observations

Volumes preserved application data and containers communicated successfully through Docker networking.

Conclusion

Docker storage and networking features enabled creation of practical multi-container applications.

Viva Preparation

1. What is a Docker Volume?
2. Why is persistent storage required?
3. What is Docker Networking?
4. What is a Bridge Network?
5. How do containers communicate?
6. What is a Multi-Container Application?
7. Why are Docker volumes important?

Troubleshooting Box

Problem	Possible Cause
Data lost after restart	Volume not configured
Containers cannot communicate	Network configuration issue
Application cannot access database	Network connectivity problem
Volume mount failure	Incorrect volume mapping

Industry Relevance

Almost every production containerized application uses persistent storage and container networking. Understanding these concepts is essential before moving to Docker Compose and Kubernetes.

Session Summary

Students learned how Docker volumes preserve application data, how Docker networks enable communication between services, and how multi-container applications operate in real-world environments.